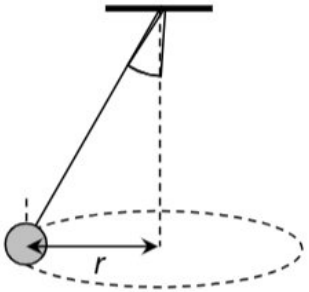
9 A metal bob is suspended with a light inextensible cord from the ceiling. It is set to whirl in a horizontal circle of radius r with a constant speed such that the cord is inclined at an angle θ to the vertical as shown in the diagram.



Determine the period of rotation for the metal bob.

A $\sqrt{\frac{4\pi^2 r}{g}}$ **B** $\sqrt{\frac{4\pi^2 r}{g sin sin\,\theta}}$ **C** $\sqrt{\frac{4\pi^2 r}{g tan tan\,\theta}}$ **D** $\sqrt{\frac{4\pi^2}{gr tan tan\,\theta}}$